

h310c Diode Voltage Controlled Filter User Manual

05-2026 Rev. 1

1 Introduction

The *310c Diode Voltage Controlled Filter* is a basic VCF based on the diode ladder topology in a small 4hp format. It is a two-pole lowpass filter, meaning it has a -12 dB/oct frequency response.

- talk about low frequencies at high resonance
- topology allows squelchy resonance
- potentially something else

2 Installation & Power

This module requires 4hp of free space in a rack. When installing, always turn off the power supply before plugging anything in. When plugging in the 10-16 pin ribbon power cable, ensure that the red stripe lines up with the **RED LINE** mark (see fig. 2.1) on the module and the marking that indicates **-12v** on the bus board.

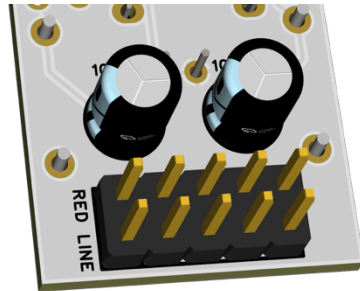


fig. 2.1 Power Connector

This module **is** reverse polarity protected. This means that the module will not get heavily damaged if the power cable is plugged in backwards. However, a reversed power cable may still cause damage to the components in the module, as well as other modules connected to the same power supply.

3 Front Panel Overview



The 310c has a common and straightforward interface similar to many VCF designs. The cutoff **FREQUENCY** knob allows for sweeping across the full frequency range (20 – 20,000 *Hz*)¹, and the **RESONANCE** knob allows for control of resonance from completely zero up to self-oscillation (at about 2-3 o'clock) and overdriving self-oscillation. The resonance intensity can be adjusted, refer to Section 4.1.

The audio **IN** and **CV** inputs both accept standard Eurorack signal levels of up to 24V_{pp}, and can be attenuated using their respective onboard attenuators, indicated by the panel. The **IN** attenuator can be used to overdrive the input signal when fully open, which will decrease the effect of resonance and result in a soft-clipped output signal. **CV** is capable of tracking 1 V/*oct* for around 2-3 octaves, depending on the attenuator position. To achieve this, the attenuator should be set at about 1-3 o'clock, the exact position also depends on the **FREQUENCY** knob position.

Finally, the jack below the **CV** input is **OUT**, which is the output of the filter.

fig. 3.1 Panel Overview

¹ Exact frequency range of the filter knob has yet to be measured

4 Additional Information

4.1 Resonance Intensity Adjustment

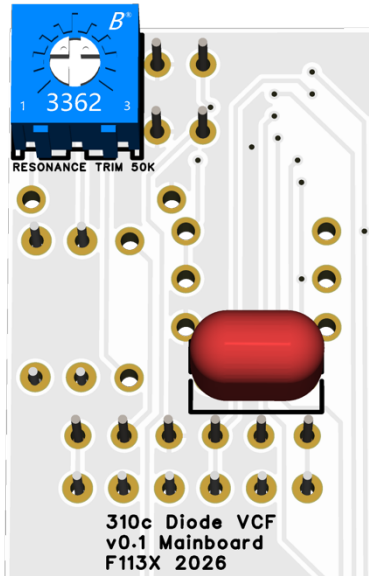


fig 4.1 Resonance Trimmer

The intensity of the resonance can be adjusted via a trimmer potentiometer at the bottom PCB of the module. To adjust it, First turn the **FREQUENCY** knob to the central (12 o'clock) position and the **RESONANCE** knob to the maximum position, and plug a cable to **OUT** to monitor the output of the module. Next, remove the module from the rack and locate the resonance trimmer on the bottom PCB (see fig. 4.1). Using a small flathead screwdriver, adjust the frequency trimmer to your liking. The output of the resonance will be the resonance of the module with maximum resonance.

Warning:

- a. High resonance intensity can result in very high levels. Be careful to not damage hearing or speakers.
- b. Touching the PCB while adjusting the trimmer **will** cause instabilities in the module. While it won't cause damage in the module, but will make adjustment more difficult. It is recommended to hold the panel or place the module upside-down on the rack (see fig 4.2) while adjusting.
- c. **Do not** allow contact from any metal with the PCB, as it may cause damage to the circuit.

4.2 Frequency Response

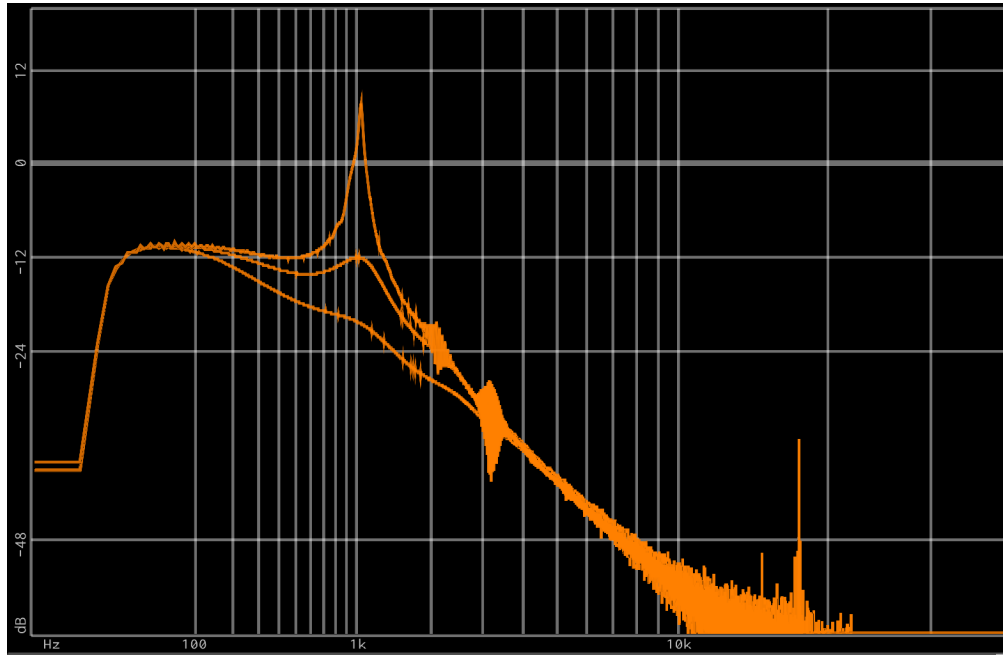


fig 4.2 Frequency Response Diagram

Fig 4.2 shows the frequency response of the filter measured by the Bogaudio Ranalyzer (VCVRack). The **FREQUENCY** knob was set at noon, which had a cutoff frequency of around 1.05kHz . The 3 contour lines show 3 different resonance settings, which are, from low to high, 0 resonance (**RESONANCE** fully counterclockwise), medium resonance, and high resonance (right before self-oscillation)².

4.3 Self-Oscillation

² Medium resonance and high resonance knob positions depend on trimmer settings

5 Patch Ideas

As this module is a very basic module without complicated functionality, creative uses for it are rather limited other than the standard VCF.

5.1 Filter Pinging

When triggers are patched into the filter while the **RESONANCE** is right before self-oscillation, the filter can be pinged, outputting a short decaying sine pulse. The frequency of the ping is dependent on the cutoff **FREQUENCY**, and the decay time can be slightly adjusted using the **RESONANCE** control.

CV can be used to obtain more interesting results, try patching a simple decaying envelope to it to obtain a pitch sweep, allowing the synthesis of kick drums or other percussive sounds. When an audio rate signal is patched into **CV**, more complex sounds can be obtained, reminiscent of *Buchla Bongos* and the likes.

Experiment with the combinations of **IN** and **CV** inputs, for example a decaying envelope into **IN** and a trigger into **CV**. Also, experiment with combining different signals in an external mixer to obtain complex modulation.

5.2 Sine VCO

When **RESONANCE** is pushed into self-oscillation, it outputs a sine wave with relatively low distortion³. Therefore, the **CV** input can be used as a frequency modulation input of sorts, and when the **CV** attenuator is adjusted correctly, the **CV** input can be used for 1 V/oct tracking for a usable range (see Section 3). Thus, the filter can be used as a sine VCO.

Experiment with wavefolding the sine wave, using it to modulate other controls, overdriving the signal, as they will all result in interesting results. Additionally, driving the initial signal by cranking **RESONANCE** will produce an output of higher harmonic content, try filtering the driven output for other results. Note that the signal is hard-clipped, and may produce noisy and/or loud results.

³ Exact distortion of the sine waves has yet to be measured.

6 Technical Specifications

6.1 Dimensions & Power

Width	Depth	+12v Current	-12v Current
<i>8hp</i>	<i>~35mm max.</i>		

6.2 Inputs & Outputs

Inputs		Outputs	
IN	<i>Any modular signal levels</i>	OUT	<i>~20 V_{pp} max.</i>
CV			

6.3 Other Specifications